Experiment – 01

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Semester: 5th

Branch: BE-CSE(AIML)

Subject: Advanced Database Management System

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

Assumption:

- There are no duplicates in the **candidates** table.

candidates Table:

Column Name	Type
candidate_id	integer
skill	varchar

candidates Example Input:

candidate_id	skill
123	Python
123	Tableau
123	PostgreSQL
234	R

candidate_id	skill
234	PowerBI
234	SQL Server
345	Python
345	Tableau

Example Output:

candidate_id
123

```
SELECT candidate_id FROM candidates WHERE skill = 'Python'
INTERSECT
SELECT candidate_id FROM candidates WHERE skill = 'Tableau'
INTERSECT
SELECT candidate_id FROM candidates WHERE skill = 'PostgreSQL'
ORDER BY 1;
```

Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

pages Table:

Column Name	Type
page_id	integer
page_name	varchar

pages Example Input:

page_id	**page_name**
20001	SQL Solutions
20045	Brain Exercises
20701	Tips for Data Analysts

page_likes Table:

Column Name	**Type**
user_id	integer
page_id	integer
liked_date	datetime

page_likes Example Input:

user_id	**page_id**	**liked_date**
111	20001	04/08/2022 00:00:00
121	20045	03/12/2022 00:00:00
156	20001	07/25/2022 00:00:00

Example Output:

page_id
20701

```
SELECT page_id FROM pages
WHERE page_id NOT IN (SELECT page_id FROM page_likes)
```

Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- **parts_assembly** table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a **finish_date**.

This question is straightforward, so let's approach it with simplicity in both thinking and solution.

Effective April 11th 2023, the problem statement and assumptions were updated to enhance clarity.

parts_assembly Table

Column Name	Type
part	string
finish_date	datetime
assembly_step	integer

parts_assembly Example Input

part	finish_date	assembly_step
battery	01/22/2022 00:00:00	1
battery	02/22/2022 00:00:00	2
battery	03/22/2022 00:00:00	3
bumper	01/22/2022 00:00:00	1
bumper	02/22/2022 00:00:00	2
bumper		3

part	finish_date	assembly_step
bumper		4

Example Output

part	assembly_step
bumper	3
bumper	4

```
SELECT
  DISTINCT(part),
  assembly_step
FROM Parts_assembly
WHERE finish_date IS NULL;
```